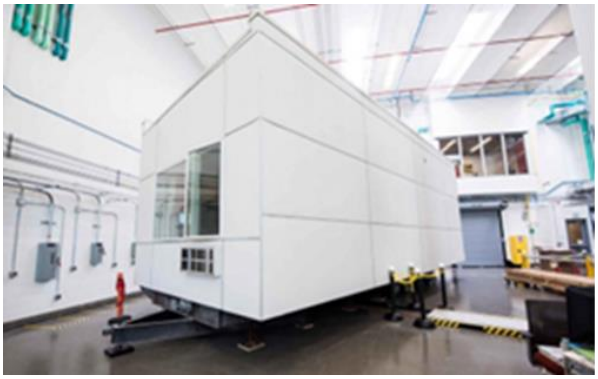
Project 1: EnergyPlus/OpenStudio Modeling of iUnit

1. Project Description

The goal of Project 1 is to create EnergyPlus/OpenStudio Model of iUnit based on the information and measurement given in this document. iUnit is a modular apartment located at the campus of National Renewable Energy Laboratory (NREL) that can be moved indoor for guarded testing or outdoor for environmental testing. NREL is using iUnit to conduct validation-grade energy simulation experiments.





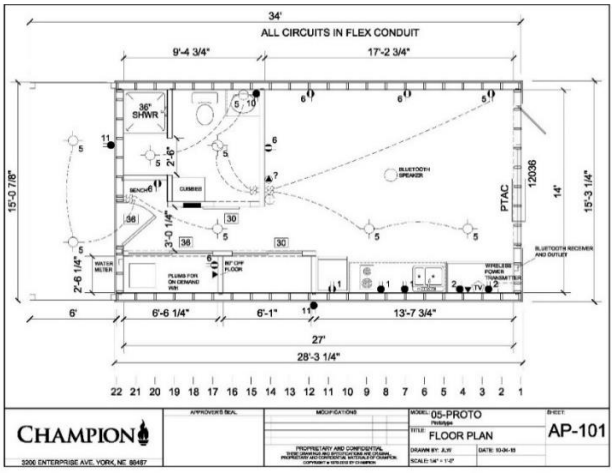
You will use OpenStudio For the geometry modeling of iUnit. For the rest part of the modeling, you can choose whether to use EnergyPlus or OpenStudio in this project. The benefits of using OpenStudio are 1) use one software to finish the project, and 2) a better user interface. The benefits of using EnergyPlus are 1) no need to self-teach OpenStudio, 2) everything related to EnergyPlus is covered in class, and 3) easier HVAC template creation. Overall, EnergyPlus is suggested in Project 1.

2. iUnit Details

2.1. Building Geometry

- Use FloorspaceJS in OpenStudio to create building geometry model based on the rest of the content in ***Section 2.1***. The tutorial on using FloorspaceJS in OpenStudio to create building geometry, thermal zones, and subsurfaces is shown in the following two links.
  - Create Geometry: https://www.youtube.com/watch?v=-5HHfgSlTsc
  - Add Thermal Zones and Subsurfaces: https://www.youtube.com/watch?v=22Lj6kdwWrc
- The floor plan and the matched internal space modeling are shown in Figure 1. The floor plan drawing can be found in **Appendix A**. You only need to model the exterior walls, interior walls (excluding interior door), one window, and one exterior door, and ignore the other elements shown in Figure 1.
    - The height of ceiling is 11.01 ft.
    - Only model the internal wall shown as the dotted lines in Figure *1* (b)

(a)



(b)

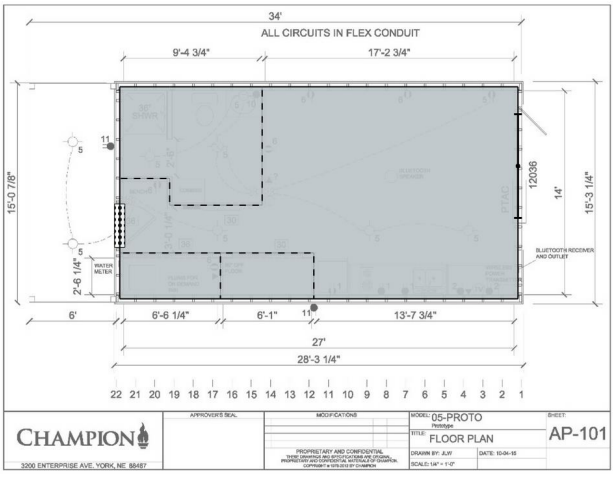


Figure 1. Use FloorspaceJS to create model geometry

- Door dimension and position
  - Door dimension is based on the midline dimension described below:
    - Door Height (ft): 7.27
    - Door Width (ft): 3.00
  - Door position is shown in the figure below (red circle).

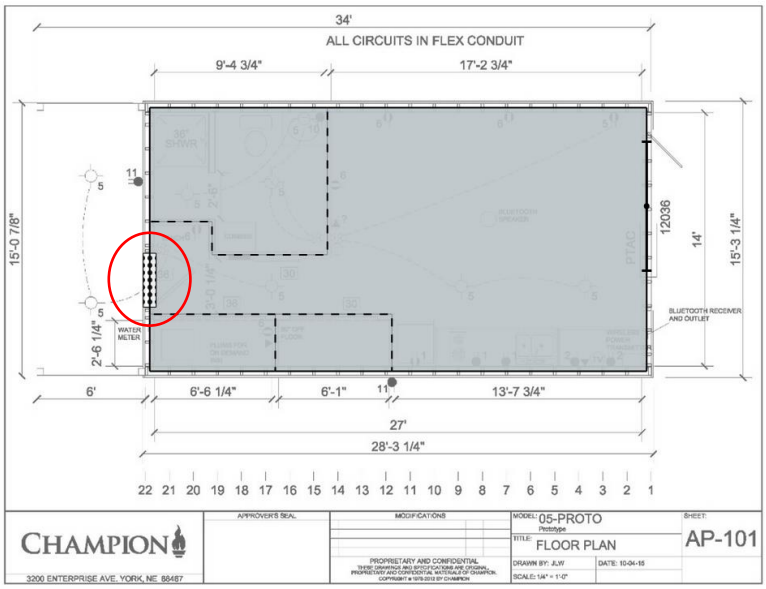


Figure 2. Door position shown in FloorspaceJS

- Window dimension and position
  - Window width: 9.000 ft
  - Window height: 4.917 ft
  - The sill height is 2.988 ft
  - The position (from left) of the window is: 3.448 ft
  - The position (from right) of the window is: 2.114 ft

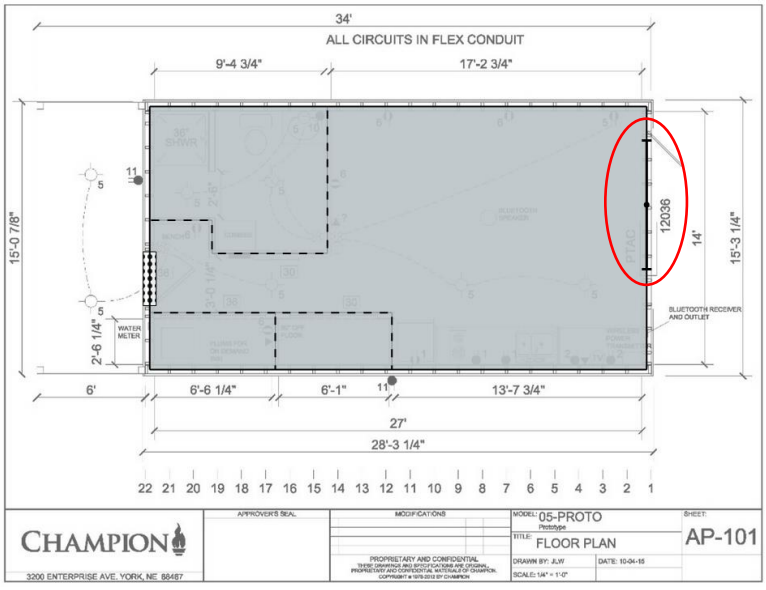


Figure 3. Window position shown in FloorspaceJS

- Orientation
  Assume the orientation by yourself, since iUnit is a mobile house.
- Thermal zones
  Create thermal zones based on the diagram below.

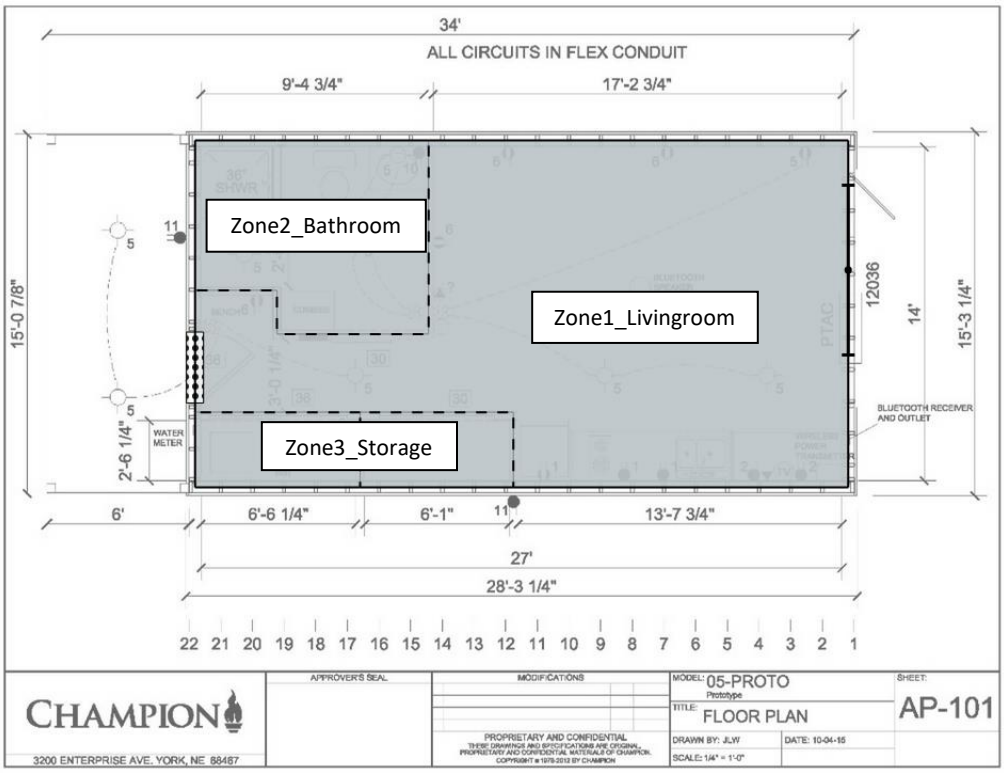


Figure 4. Thermal zone diagram

There are three thermal zones in iUnit.

- Zone1_Livingroom
- Zone2_Bathroom
- Zone3_Storage

- The above numbers are all based on measurements. If there are conflicting measurements, the modeler can make assumptions by themselves to allow certain levels of errors.
- The modeled geometry in OpenStudio should look like the following figures.

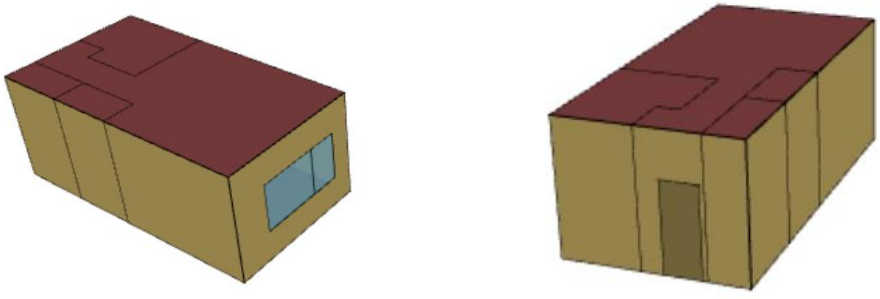


Figure 5. The appearance of finished geometry model in OpenStudio. Left: showing window; Right: showing door.

- If you plan to use EnergyPlus to finish Project 1, export the whole osm file with the completed geometry to idf file shown below, and finish the rest modeling in EnergyPlus. If you plan to use OpenStudio to finish Project 1, just keep finishing the rest modeling in this osm file.

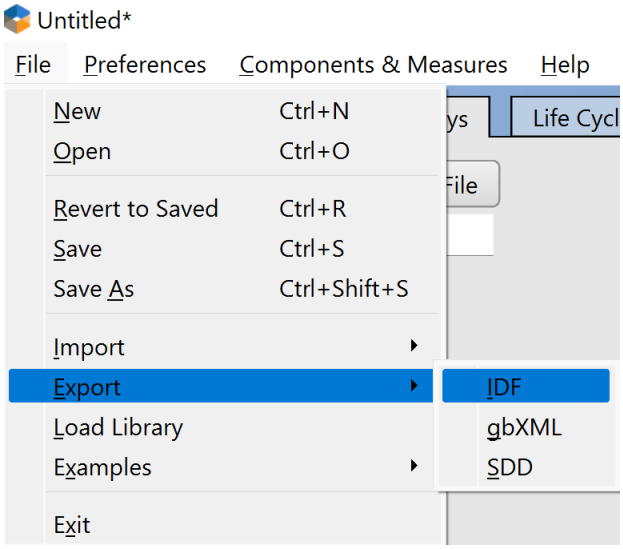


Figure 6. Export the geometry to idf file

2.2. Weather File

- NREL only measured weather data from 1/1/2021 to 4/25/2021. So the running period should be 1/1/2021 to 4/25/2021 defined in the object "RunPeriod"
- Create "epw" weather file based on the weather data in "csv" format in Appendix B, using the built-in software "Weather Statistics and Conversions" to convert csv to epw weather file.

2.3. Constructions

Hint 1: Use the following objects to define constructions:

- Material (for walls and doors)
- WindowMaterial:SimpleGlazingSystem (for window)
- Construction
- BuildingSurface:Detailed (for walls)
- FenestrationSurface:Detailed (for window and doors)

Hint 2: You can refer to and imitate the way the reference building (**Appendix D**) models materials and construction on each surface.

Hint 3: You can switch SI to imperial unit in EnergyPlus by clicking the "View" tab

2.3.1. Floor

Table 1. Details on floor construction layers

Floor Layer 1 (Outside): Rodent Barrier (HDPE)		
Items	Value	Unit
Roughness	Smooth	NA
Thickness	0.05	in

Conductivity	3.472	Btu·in/hr·ft^2·R
Density	59.3	lb/ft^3
Specific Heat	0.4347	Btu/lb·R
Floor Layer 2: Insulated Joist Layer		
Items	Value	Unit
Roughness	Smooth	NA
Thickness	14.462	in
Conductivity	0.3323	Btu·in/hr·ft^2·R
Density	6.336	lb/ft^3
Specific Heat	0.2357	Btu/lb·R
Floor Layer 3: OSB		
Items	Value	Unit
Roughness	Smooth	NA
Thickness	1.75	in
Conductivity	0.826	Btu·in/hr·ft^2·R
Density	33.96	lb/ft^3
Specific Heat	0.289	Btu/lb·R
Floor Layer 4 (Inside): Plywood		
Items	Value	Unit
Roughness	Smooth	NA
Thickness	0.25	in
Conductivity	0.8	Btu·in/hr·ft^2·R
Density	33.96	lb/ft^3
Specific Heat	0.289	Btu/lb·R

2.3.2. Roof

Table 2. Details on roof construction layers

Roof Layer 1 (Outside): EPDM Rubber		
Items	Value	Unit
Roughness	Smooth	NA
Thickness	0.1	in
Conductivity	1.3889	Btu·in/hr·ft^2·R
Density	63	lb/ft^3
Specific Heat	0.48	Btu/lb·R
Solar Absorptance	0.3	NA
Roof Layer 2: OSB		
Items	Value	Unit
Roughness	Smooth	NA
Thickness	0.5	in
Conductivity	0.8264	Btu·in/hr·ft^2·R
Density	33.96	lb/ft^3
Specific Heat	0.289	Btu/lb·R
Roof Layer 3: Insulted Truss Layer		
Items	Value	Unit
Roughness	Smooth	NA
Thickness	37.375	in
Conductivity	0.3425	Btu·in/hr·ft^2·R
Density	3.296	lb/ft^3
Specific Heat	0.1900	Btu/lb·R
Roof Layer 4 (Inside): Drywall		

Items	Value	Unit
Roughness	MediumSmooth	NA
Thickness	0.625	in
Conductivity	1.1111	Btu·in/hr·ft^2·R
Density	49.94	lb/ft^3
Specific Heat	0.26034	Btu/lb·R

2.3.3. Exterior Wall

Table 3. Details on Exterior wall construction layers

Wall Layer 1 (Outside): Exterior Finish OSB		
Items	Value	Unit
Roughness	Smooth	NA
Thickness	0.625	in
Conductivity	0.8264	Btu·in/hr·ft^2·R
Density	33.96	lb/ft^3
Specific Heat	0.289	Btu/lb·R
Solar Absorptance	0.5	NA
Wall Layer 2: Rigid Insulation		
Items	Value	Unit
Roughness	Smooth	NA
Thickness	1	in
Conductivity	0.2	Btu·in/hr·ft^2·R
Density	2.2786	lb/ft^3
Specific Heat	0.358	Btu/lb·R
Wall Layer 3: OSB		
Items	Value	Unit
Roughness	Smooth	NA
Thickness	0.5	in
Conductivity	0.8264	Btu·in/hr·ft^2·R
Density	33.96	lb/ft^3
Specific Heat	0.289	Btu/lb·R
Wall Layer 4: Stud Layer		
Items	Value	Unit
Roughness	Smooth	NA
Thickness	5.5	in
Conductivity	0.2927	Btu·in/hr·ft^2·R
Density	7.541	lb/ft^3
Specific Heat	0.2573	Btu/lb·R
Wall Layer 5 (Inside): Drywall		
Items	Value	Unit
Roughness	Medium Smooth	NA
Thickness	0.625	in
Conductivity	1.111	Btu·in/hr·ft^2·R
Density	49.94	lb/ft^3
Specific Heat	0.26034	Btu/lb·R

2.3.4. Interior Wall

Assume interior wall material by yourself. Suggest using commonly used materials in residential constructions.

2.3.5. Window

- Use Simple Glazing System Window, U-factor = 0.223 Btu/ft2·hr·R
- Solar Heat Gain Coefficient = 0.43
- Visible Transmittance = 0.42

2.5.6. Door

- Insulated steel exterior door

Layer (Interior to exterior	Thickness (in)	Nominal R-Value per Inch (ft2·°F·h/BTU·in)	Calbrated R-Value per Inch (ft2·°F·h/BTU·in)	Total Calibrated R-Value (ft2·°F·h/BTU)
Expanded Polystyrene	1.5	4.90	4.90	7.35

Figure 7. Door materials

- Conductivity = 0.20408 BTU in/ft2 F h
- Density of Expanded Polystyrene: 24.0 kg/m^3
- Specific heat of Expanded Polystyrene: 1210.0 J/kg*K

2.4. Internal Loads

2.4.1. People

Internal load “People” strictly follows the National Renewable Energy Laboratory’s business hours, 1) Monday–Thursday, 8 a.m.–5 p.m, 2) Friday, 8 a.m.–4 p.m., 3) Closed on Saturday and Sunday. There are on average two persons in iUnit during business hours and no people in time of the rest. (hint: 1) use object “people” to define number of people, and 2) use object “schedule:compact” or “schedule:file” to define schedule)

2.4.2. Lighting

Assume lighting capacity by yourself since it is not measured. This kind of assumption of missing values is very common in EnergyPlus. Refer to ASHRAE standard. The lighting on/off schedule is assumed to be the same as the business hours in 2.3.2. (hint: 1) use object “lights” to define baseline lighting power, and 2) use object “schedule:compact” or “schedule:file” to define schedule)

2.4.3. Electric Equipment

The measured background electricity can be found in Appendix C. Make sure the EnergyPlus/OpenStudio model exactly reflects this measurement. The electric equipment load is defined by baseline and schedule. Hint 1) use object “ElectricEquipment” for baseline, 2): use object “schedule:file” for schedule; and 3) you cannot directly load Appendix C as “schedule”. Instead, you can set the “baseline” as the maximum value of the whole time, and calculate the timeseries percentages as “schedule”.

2.5 Infiltration

The infiltration will be defined using EnergyPlus object ZoneInfiltration:DesignFlowRate. In this object, the main inputs are shown in Equation 1.

$$\text{Infiltration} = (I_{design})\,(F_{schedule})\,[A + B\,|(T_{zone} - T_{odb})| + C\,(\text{WindSpeed}) + D\,(\text{Windspeed}^2)]$$ Equation 1

According to the onsite measurements, I = 1, Fschedule = always on (1,1,1,1,…), A = 0.03, B = 0.003, C = 0, D = 0. Apply the infiltration to all zones.

2.6 HVAC System

Use the HVAC template to design your own HVAC system for iUnit. Choose one from below:

- Packaged terminal air conditioner (PTAC) systems with optional hot water boiler
- Direct-expansion heat pump systems
- Packaged variable air volume systems using direct-expansion cooling

The setpoints of each zone are listed as follows

- Zone1_Livingroom is conditioned with dual cooling and heating setpoint. The cooling setpoint is 76F and the heating setpoint is 72F.
- Zone2_Bathroom is conditioned with a single heating setpoint. The heating setpoint is 74F.
- Zone3_Storage is an unconditioned room without any setpoints.

In order to size the HVAC system correctly, you need to create two objects with the type SizingPeriod:DesignDay (one for summer design day and the other for winter design day), according to which the HVAC system know the weather conditions to decide the size of the HVAC system. You can refer to the sizing period object for Phoenix, Arizona in Appendix D to create similar ones for Golden, Colorado.

3. Submission and Presentation

- Schedule

No.	Event	Time
2	Office hours for building geometry creation	9:30 AM, Nov 5, 2024
3	Office hours for the rest EnergyPlus modeling	9:30 AM, Nov 7&12, 2024
4	Final Presentation & D2L Submission Deadline	9:30 AM, Nov 14, 2024

- Action Items
    1. Follow the instructions in this document and create EnergyPlus/OpenStudio model for iUnit. Submit the finalized “IDF” file if using EnergyPlus or the “OSM” file if using OpenStudio to D2L.
    2. Conduct sensitivity analyses by choosing and manipulating two variables in your model (such as orientation, wall and window materials, HVAC performance such as COP, energy-efficient lighting, and HVAC operation schedules) that affect model energy consumption while not affecting the thermal comfort of people (indicated by variables such as unmet hours, PMV, PPD). Make PowerPoint slides to visualize the energy performance (such as hourly, weekly, monthly, and annual energy consumption, end-uses (such as lighting electricity, HVAC electricity, plug load electricity, etc.), thermal comfort, etc.) of the original model and two sensitivity analyses. Draw conclusions on how to make iUnit more energy efficient. Include a one-page slide that lists all the assumptions you made in the modeling process.

3. Additional requirement for **graduate students**: Instead of randomly manipulating two variables in your model, conduct a comprehensive sensitivity analysis of the two variables you select to evaluate their impact on building energy consumption.
4. Present your slides (10 minutes) in class.